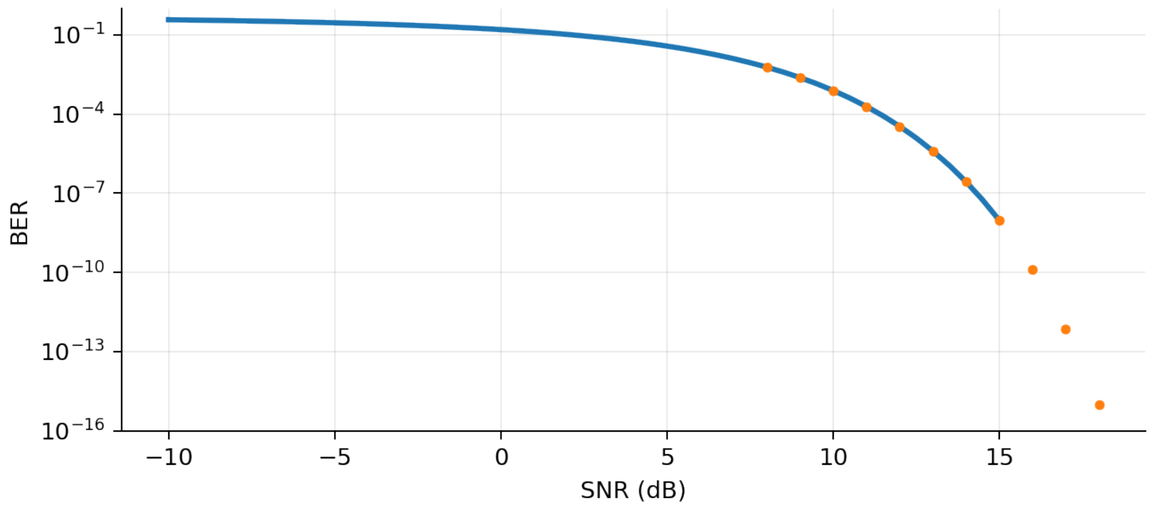
BER PERFORMANCE ANALYSIS IN DIGITAL COMMUNICATIONS

Monte Carlo Simulation and Importance Sampling

Experimental estimation of the bit error rate for bipolar NRZ signaling over an AWGN channel



UNIVERSIDAD POLITÉCNICA DE MADRID

Telecommunication Engineering - Communications Simulation Practice

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Software MATLAB

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1. Abstract and objectives

Abstract. This practice estimates the bit error rate (BER) of a binary bipolar non-return-to-zero (NRZ) link affected by additive white Gaussian noise (AWGN). A conventional Monte Carlo estimator is first compared with the analytical BER obtained from the complementary error function. The second part addresses the severe computational cost of estimating rare errors at high signal-to-noise ratio (SNR) by applying variance-inflated importance sampling. The likelihood ratio is derived and implemented so that the deliberately biased simulation remains statistically unbiased.

1.1 Objectives

- Generate an equiprobable binary sequence and map it to the bipolar levels -A and +A.
- Model an AWGN channel and implement a zero-threshold hard detector.
- Estimate BER experimentally over a wide SNR range by direct Monte Carlo simulation.
- Validate the simulation against the theoretical complementary-error-function expression.
- Estimate very small BER values efficiently through importance sampling.
- Identify implementation errors that can bias the weighted estimator and propose corrections.

1.2 Main outcome

The conventional estimator agrees with theory while the expected number of observed errors is sufficiently large. At high SNR, however, it reaches a simulation floor because an experiment with N samples cannot resolve probabilities much smaller than approximately 1/N. The importance-sampling estimator continues to follow the theoretical curve many orders of magnitude below that floor using only 10^5 samples per SNR point.

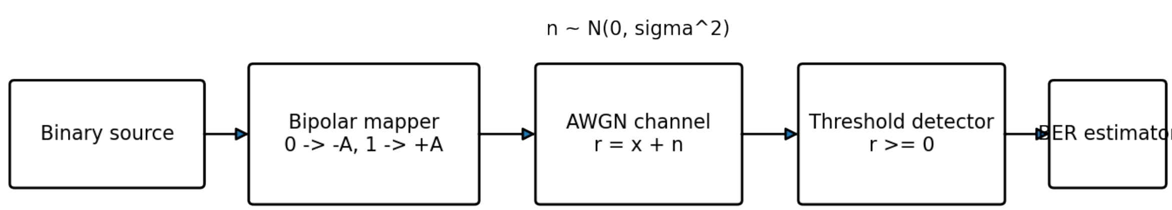

Figure 1. Baseband simulation chain used by both estimators.

2. System model and theoretical BER

2.1 Binary source and bipolar mapping

The source produces independent equiprobable bits b in {0,1}. The transmitter converts each bit into an antipodal amplitude x according to

```
x = 2A b - A  ->  x = -A for b = 0, and x = +A for b = 1.
```

The received sample is r = x + n, where n is a zero-mean Gaussian random variable with variance sigma^2. The detector decides bit 1 when r >= 0 and bit 0 otherwise.

2.2 SNR convention used by the scripts

The MATLAB expressions define the sample SNR as A^2/sigma^2. Therefore, in decibels, SNR_dB = 20 log10(A/sigma), and the required noise standard deviation is

```
sigma = A * 10^(-SNR_dB/20).
```

This convention is internally consistent with the theoretical BER used in the program. In a complete pulse-shaped communication system, the relationship with Eb/N0 additionally depends on pulse energy and receiver normalization.

2.3 Analytical error probability

For either transmitted symbol, an error occurs when Gaussian noise crosses a boundary located A units away from the symbol. By symmetry, the bit error probability is

```
P_b = Q(A/sigma) = 0.5 * erfc( A / (sqrt(2)*sigma) ).
```

This expression is used as the reference curve against which both numerical estimators are evaluated.

Quantity	Meaning	MATLAB expression
A	Bipolar signal amplitude	A = 1
sigma	Noise standard deviation	A.*10.^(-SNR/20)
r	Received sample	x + noise
Decision	Hard threshold at zero	rx >= 0
BER theory	Gaussian tail probability	erfc(A./(sqrt(2)*sigma))/2

3. Conventional Monte Carlo method

3.1 Estimator

For each SNR value, N independent bits are transmitted, detected and compared with the original sequence. Let I_k be 1 when the kth bit is detected incorrectly and 0 otherwise. The Monte Carlo estimator is the sample mean

```
BER_hat_MC = (1/N) * sum(I_k),   k = 1,...,N.
```

Because I_k is Bernoulli with probability p, the estimator is unbiased and has variance p(1-p)/N. Its approximate relative standard deviation is 1/sqrt(Np) for small p. Consequently, a stable estimate normally requires tens or hundreds of observed errors, not merely one.

3.2 Rare-event limitation

At N = 10^7, a BER of 10^-6 produces an average of ten errors and is already noisy. A BER of 10^-8 produces only 0.1 expected errors, so most runs return zero. Zero observed errors does not prove that the true BER is zero; it only indicates that the simulation is too short to resolve the event.

True BER	Expected errors at N=10^7	Typical behavior	Interpretation
10^-3	10,000	Very stable	Direct MC is efficient
10^-5	100	Usable	Moderate statistical spread
10^-7	1	Highly variable	Many runs give 0 or 1 error
10^-9	0.01	Almost always zero	Direct MC is impractical

3.3 Memory-safe implementation

The original vectorized form allocates several arrays containing 10 million samples. Since each double-precision array requires about 80 MB, peak memory can become unnecessarily high. The delivered script processes blocks of 2 x 10^5 samples, accumulates the number of errors and preserves exactly the same estimator.

```
while processed < N_MC
    currentBlock = min(blockSize, N_MC - processed);
    data = rand(1,currentBlock) >= 0.5;
    tx = 2*A*double(data) - A;
    rx = tx + sigma_MC(i)*randn(1,currentBlock);
    errorCount = errorCount + nnz(xor(rx >= 0, data));
    processed = processed + currentBlock;
end
```

4. Importance sampling

4.1 Principle

Importance sampling deliberately generates samples from an alternative probability density that visits the error region more often. Each observation is then multiplied by a likelihood ratio so that the final mean represents the original channel rather than the modified simulation.

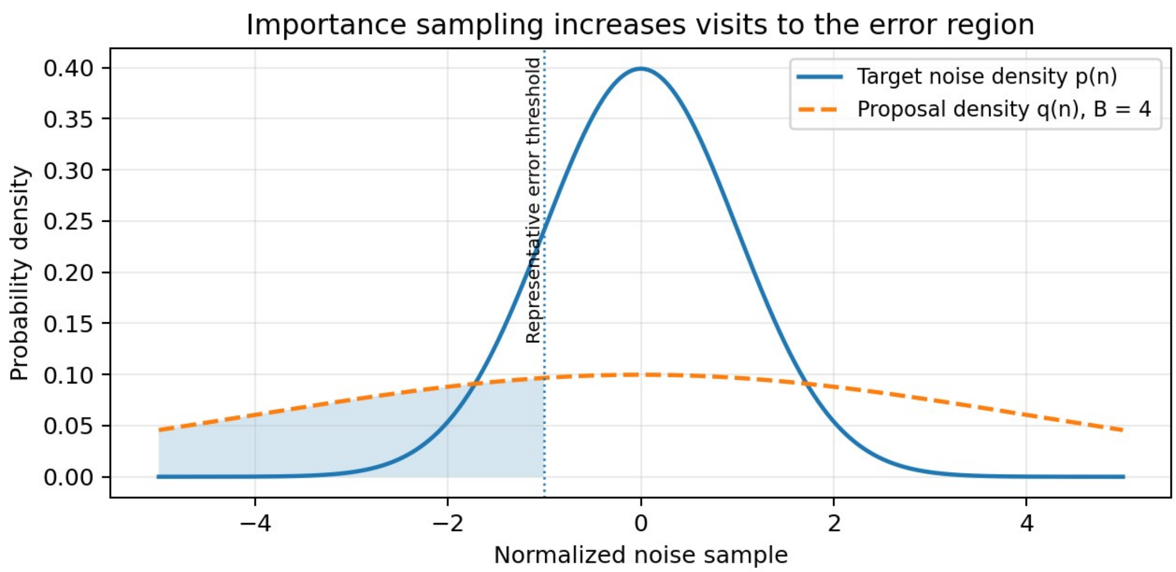

Figure 2. The broader proposal distribution creates more threshold crossings.

4.2 Proposal distribution and likelihood ratio

The true noise density is Gaussian with standard deviation sigma. The proposal density uses the larger standard deviation B sigma, with B = 4 in the practice. Denoting the true density by p(n) and the proposal by q(n), the required weight is

```
w(n) = p(n)/q(n)
     = B * exp{ -n^2/(2 sigma^2) * [1 - 1/B^2] }.
```

The unbiased importance-sampling estimator is therefore

```
BER_hat_IS = (1/N) * sum( w(n_k) * I_k ).
```

4.3 Critical correction to the supplied code

In the supplied fragment, the instruction errores = sum(xor(rxd,data),1) converts the complete error-indicator vector into a single scalar before weighting. Multiplying that scalar by every likelihood weight does not implement the estimator above and produces a biased result. The corrected implementation keeps one logical indicator per sample:

```
errorIndicator = xor(detected, data);
BER_IS(i) = mean(weight .* double(errorIndicator));
```

4.4 Choice of the scale factor B

A value B > 1 increases the frequency of errors. If B is too close to 1, the gain is small. If it is excessively large, most samples receive extremely small weights and estimator variance can increase again. The value B = 4 is a reasonable laboratory setting for the SNR range from 8 to 18 dB, but it is not universally optimal.

5. MATLAB implementation

5.1 File MonteCarloYEnfatizado.m

This is the complete version of the practice. It contains the two estimators, theoretical curves, timing measurements, figures, standard-error computation for importance sampling and export of all variables to BER_results.mat.

Instruction	Purpose
rng(42, twister)	Makes the numerical experiment reproducible.
rand >= 0.5	Creates equiprobable logical bits.
2*A*double(data)-A	Maps bits to bipolar NRZ amplitudes.
randn	Generates standard Gaussian samples.
nnz(xor(...))	Counts hard-decision errors efficiently.
erfc	Evaluates the exact Gaussian-tail BER.
semilogy	Displays probabilities spanning many decades.
save	Stores the simulation arrays and timing variables.

5.2 File MCandIIS.m

This second script follows the compact structure of the original laboratory template. It is useful for direct comparison with the supplied material, but it incorporates the same two essential corrections: a memory-safe block loop for N = 10^7 and sample-by-sample weighting in the importance-sampling estimator.

5.3 Plotting corrections

The supplied code plots the experimental curve in white (-w). On MATLAB figures with the default white background, that curve can become invisible. The delivered scripts use visible blue and green curves and include explicit titles, axis labels, grid lines and legends.

5.4 Reproducibility and execution time

The random seed is fixed so that repeated executions on the same MATLAB version produce consistent outputs. Exact execution time is hardware-dependent. The conventional experiment is intentionally much slower because it processes 10^7 bits at every one of 51 SNR points, whereas the emphasized simulation uses 10^5 samples at 11 points.

6. Results and discussion

6.1 Conventional Monte Carlo results

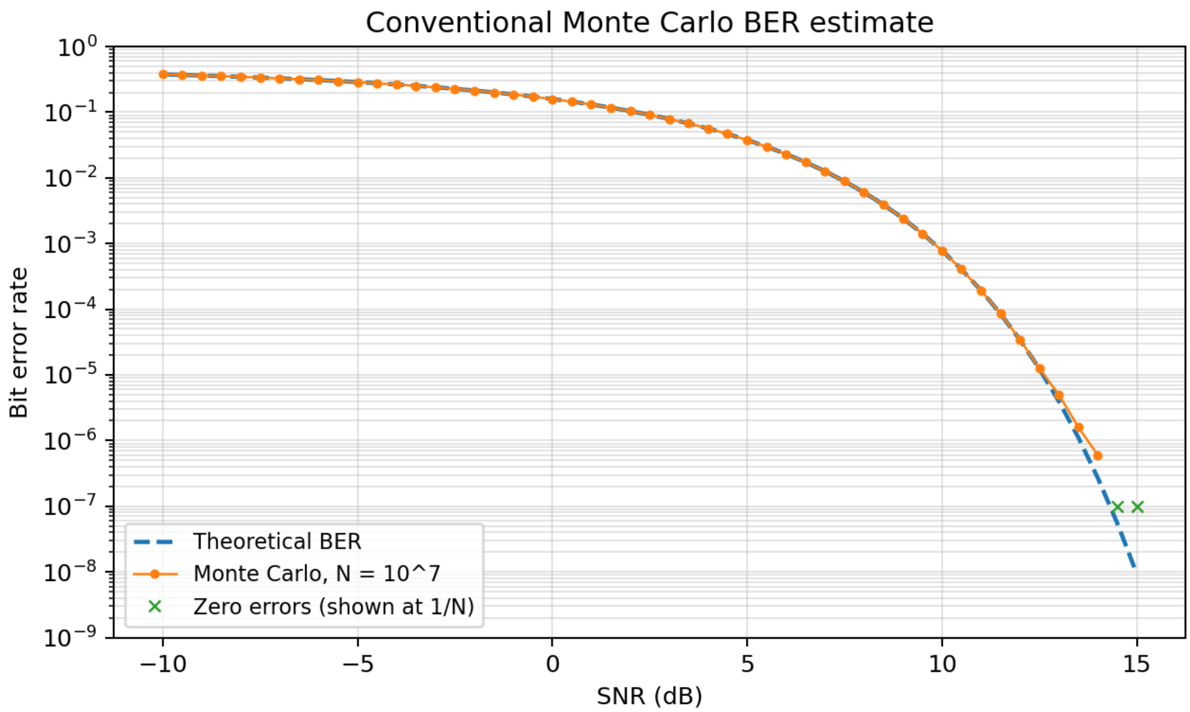

Figure 3. Direct simulation agrees with theory until the finite-sample error floor is reached.

At low and medium SNR, the simulated points closely follow the analytical curve because each run contains many error events. In the high-SNR region, the number of observed errors decreases rapidly. Points with zero errors are displayed at 1/N only to make their location visible on a logarithmic plot; they are not genuine BER estimates equal to 10^-7.

SNR (dB)	Theoretical BER	MC estimate	Observed errors
-10.0	3.759e-01	3.758e-01	3,758,407
-5.0	2.869e-01	2.869e-01	2,868,511
0.0	1.587e-01	1.585e-01	1,585,242
5.0	3.768e-02	3.762e-02	376,191
10.0	7.827e-04	7.782e-04	7,782
15.0	9.361e-09	0.000e+00	0

6.2 Importance-sampling results

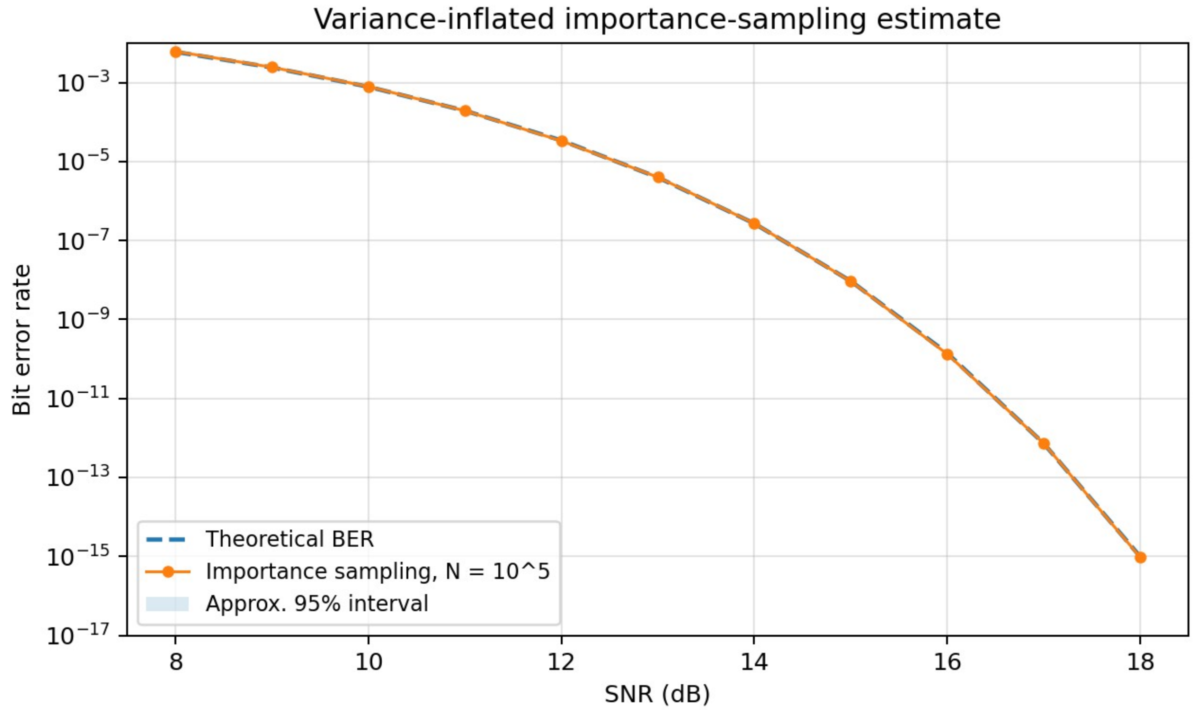

Figure 4. Importance sampling follows the theoretical BER deep into the rare-event region.

With only 10^5 proposal samples per SNR point, the weighted estimator remains close to theory over probabilities that are many orders of magnitude below the resolution of the direct simulation. The shaded band is an approximate 95% interval computed from the sample variance of the weighted error contributions.

SNR (dB)	Theoretical BER	IS estimate	Relative error	Approx. 95% half-width
8	6.004e-03	6.154e-03	2.5%	1.60e-04
9	2.413e-03	2.449e-03	1.5%	6.87e-05
10	7.827e-04	7.931e-04	1.3%	2.42e-05
11	1.940e-04	1.936e-04	0.2%	6.51e-06
12	3.430e-05	3.345e-05	2.5%	1.24e-06
13	3.969e-06	4.019e-06	1.2%	1.67e-07
14	2.695e-07	2.704e-07	0.3%	1.29e-08
15	9.361e-09	9.285e-09	0.8%	5.14e-10
16	1.399e-10	1.330e-10	4.9%	9.10e-12
17	7.236e-13	7.279e-13	0.6%	5.98e-14
18	9.845e-16	9.511e-16	3.4%	1.05e-16

6.3 Comparative assessment

- Accuracy: both methods are unbiased when correctly implemented, but importance sampling produces much lower variance in the selected rare-event range.
- Efficiency: direct Monte Carlo spends almost all high-SNR samples confirming correct decisions; the proposal distribution reallocates simulation effort toward threshold crossings.
- Robustness: the likelihood ratio must be evaluated for every sample and multiplied by the corresponding error indicator. A scalar error count cannot be weighted afterwards.
- Numerical behavior: logarithmic axes are essential, and zero-error points must be interpreted as unresolved probabilities rather than exact zeros.
- Model scope: the experiment represents one sample per bit with an ideal threshold detector. Synchronization errors, intersymbol interference, fading and coding are outside the present model.

7. Conclusions

The practice verifies the theoretical BER of bipolar NRZ signaling in AWGN and demonstrates the fundamental rare-event limitation of conventional Monte Carlo simulation. With N = 10^7, direct simulation is reliable only while a sufficient number of errors is observed. Variance-inflated importance sampling, combined with the exact likelihood ratio, extends the measurable range to extremely small BER values using N = 10^5 samples. The delivered MATLAB files correct the weighting operation, improve figure visibility, control memory consumption and save the results for later analysis.

8. References

1. J. G. Proakis and M. Salehi, Digital Communications, 5th ed., McGraw-Hill, 2008.
2. M. C. Jeruchim, P. Balaban and K. S. Shanmugan, Simulation of Communication Systems, 2nd ed., Springer, 2000.
3. S. M. Kay, Fundamentals of Statistical Signal Processing, Volume II: Detection Theory, Prentice Hall, 1998.

Appendix A. Files included in the ZIP package

File	Description
BER_Performance_Analysis_Report.pdf	English report describing theory, implementation, corrections and results.
MonteCarloYEnfatizado.m	Complete MATLAB practice with direct MC and importance sampling.
MCandIIS.m	Compact MATLAB version based on the supplied laboratory structure.
BER_results.mat	MATLAB data file containing the variables used to prepare the figures.
Figure_1_MonteCarlo.png	Conventional Monte Carlo BER plot.
Figure_2_ImportanceSampling.png	Importance-sampling BER plot.
README.txt	Execution instructions and package summary.

Appendix B. Verification checklist

- Theoretical expression uses the same SNR convention as the simulated noise standard deviation.
- The bipolar mapping is 0 -> -A and 1 -> +A.
- The detector threshold is zero.
- The direct estimator divides the total number of bit errors by N.
- The importance-sampling estimator applies one likelihood weight to each error indicator.
- All BER plots use a logarithmic vertical axis.
- The large direct simulation is processed in blocks to reduce peak memory usage.